ONLINE RESERVATION SYSTEM

1. Title

ONLINE RESERVATION SYSTEM

Oasis Infobyte – Java Development Internship

2. Introduction

The Online Reservation System is a Java-based application designed to manage train/transport reservations. The system allows users to log in, enter passenger and journey details, book tickets, generate a unique PNR number, and cancel bookings using the PNR number.

3. Objective

Provide a secure login system.

Allow users to book train tickets.

Generate a unique PNR number.

Display booking confirmation.

Allow cancellation using PNR.

Validate user inputs.

Store reservation information using a database.

4. Technologies Used

Programming Language: Java

GUI: Java Swing

Database: SQLite / MySQL

Database Connectivity: JDBC

IDE: NetBeans / IntelliJ IDEA / Eclipse

5. Main Features

Login

Username and password.

Invalid login → access denied.

Reservation

Passenger name

Train number

Train name

Class type

Journey date

Source station

Destination station

Booking

Insert booking into database.

Generate unique PNR.

Display confirmation.

Cancellation

Enter PNR.

Fetch booking details.

Confirm cancellation.

Remove booking from database.

Validation

Required fields cannot be empty.

Train number must be numeric.

Date should be valid.

6. How the Project Works

START

↓

LOGIN FORM

↓

VALID USERNAME & PASSWORD?

↓ YES

RESERVATION MENU

↓

ENTER PASSENGER DETAILS

↓

VALIDATE INPUT

↓

BOOK TICKET

↓

GENERATE UNIQUE PNR

↓

SAVE TO DATABASE

↓

SHOW CONFIRMATION

↓

CANCEL BOOKING?

↓ YES

ENTER PNR

↓

FETCH BOOKING DETAILS

↓

CONFIRM CANCELLATION

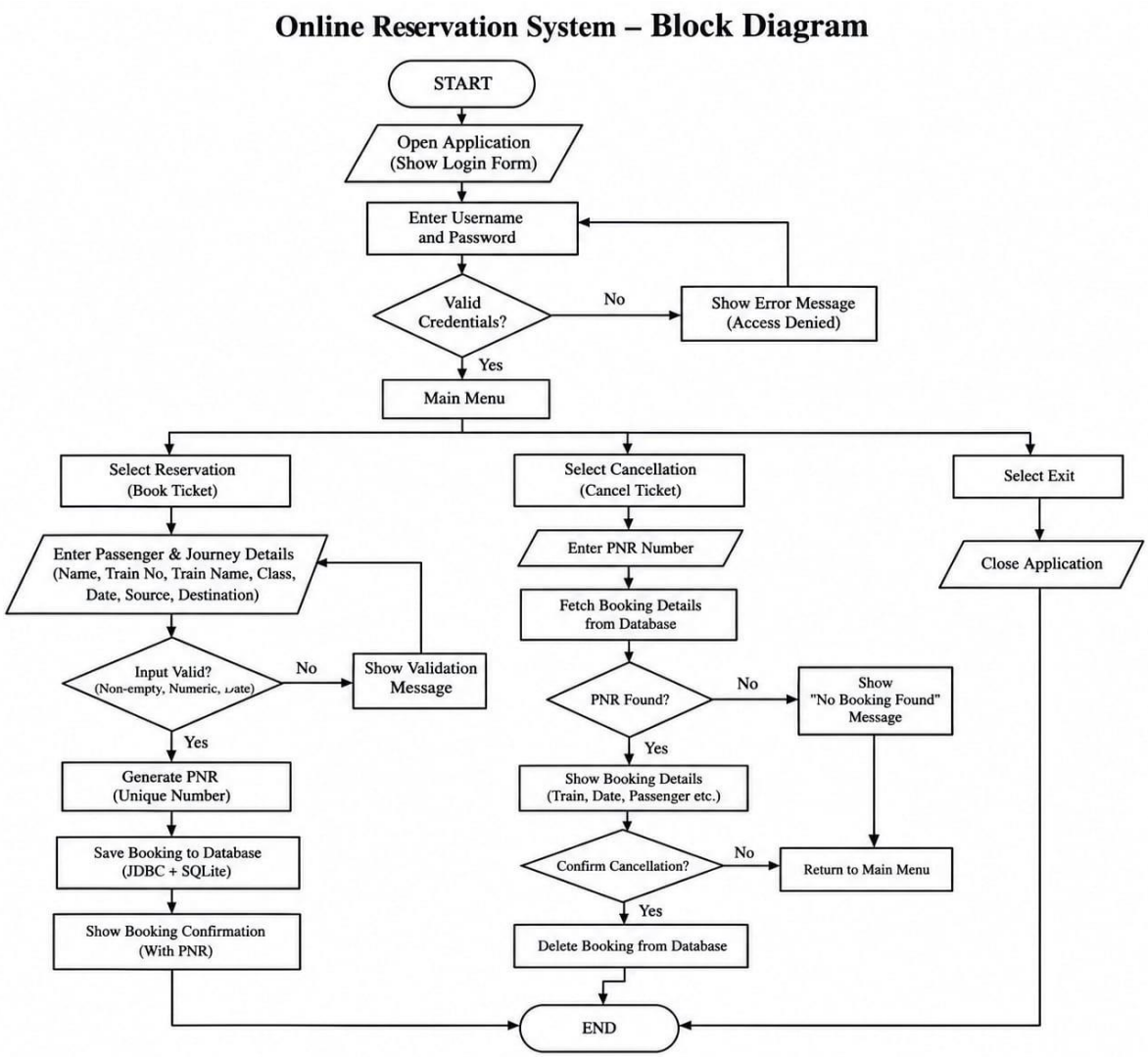
↓

DELETE FROM DATABASE

↓

END

7. Block Diagram



8. Database Table

You can create a table like:

Field

Type

PNR

INTEGER

Passenger Name

TEXT

Train Number

INTEGER

Train Name

TEXT

Class

TEXT

Journey Date

TEXT

Source

TEXT

Destination

TEXT

9.coding

```
import javax.swing.*;
import java.awt.*;
import java.sql.*;
import java.text.SimpleDateFormat;
import java.util.Date;
public class OnlineReservationSystem {
```

```
// SQLite Database
private static final String DB_URL = "jdbc:sqlite:reservation.db";
// Database Connection
public static Connection getConnection() throws SQLException {
    return DriverManager.getConnection(DB_URL);
}
// Create Database Table
public static void createTable() {
    String sql = """
            CREATE TABLE IF NOT EXISTS reservations (
                pnr INTEGER PRIMARY KEY,
                passenger_name TEXT NOT NULL,
                train_number INTEGER NOT NULL,
                train_name TEXT NOT NULL,
                class_type TEXT NOT NULL,
                journey_date TEXT NOT NULL,
                source TEXT NOT NULL,
                destination TEXT NOT NULL
            )
            """;
    try (Connection con = getConnection();
         Statement stmt = con.createStatement()) {
        stmt.execute(sql);
    } catch (SQLException e) {
        JOptionPane.showMessageDialog(null,
                "Database Error: " + e.getMessage());
    }
}
```

```
// Generate PNR
public static int generatePNR() {
    return (int) (100000 + Math.random() * 900000);
}
// Login Window
public static void loginWindow() {
    JFrame frame = new JFrame("Online Reservation System - Login");
    JLabel title = new JLabel("ONLINE RESERVATION SYSTEM");
    title.setFont(new Font("Arial", Font.BOLD, 20));
    JLabel userLabel = new JLabel("Username:");
    JLabel passLabel = new JLabel("Password:");
    JTextField username = new JTextField();
    JPasswordField password = new JPasswordField();
    JButton loginButton = new JButton("Login");
    frame.setLayout(new GridLayout(5, 2, 10, 10));
    frame.add(title);
    frame.add(new JLabel());
    frame.add(userLabel);
    frame.add(username);
    frame.add(passLabel);
    frame.add(password);
    frame.add(new JLabel());
    frame.add(loginButton);
    loginButton.addActionListener(e -> {
        String user = username.getText();
        String pass = new String(password.getPassword());
        if (user.equals("admin") && pass.equals("1234")) {
            JOptionPane.showMessageDialog(frame,
                    "Login Successful!");
```

```
            frame.dispose();
            reservationWindow();
        } else {
            JOptionPane.showMessageDialog(frame,
                    "Invalid Username or Password!",
                    "Login Failed",
                    JOptionPane.ERROR_MESSAGE);
        }
    });
    frame.setSize(500, 300);
    frame.setLocationRelativeTo(null);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setVisible(true);
}
// Reservation Window
public static void reservationWindow() {
    JFrame frame = new JFrame("Online Reservation System");
    JLabel title = new JLabel("TRAIN RESERVATION");
    title.setFont(new Font("Arial", Font.BOLD, 20));
    JLabel nameLabel = new JLabel("Passenger Name:");
    JLabel trainNoLabel = new JLabel("Train Number:");
    JLabel trainNameLabel = new JLabel("Train Name:");
    JLabel classLabel = new JLabel("Class:");
    JLabel dateLabel = new JLabel("Journey Date:");
    JLabel sourceLabel = new JLabel("Source:");
    JLabel destinationLabel = new JLabel("Destination:");
    JTextField nameField = new JTextField();
    JTextField trainNoField = new JTextField();
```

```
JTextField trainNameField = new JTextField();
JTextField dateField = new JTextField();
JTextField sourceField = new JTextField();
JTextField destinationField = new JTextField();
String[] classes = {
        "AC",
        "Sleeper",
        "First Class",
        "Second Class"
};
JComboBox<String> classBox = new JComboBox<>(classes);
JButton bookButton = new JButton("Book Ticket");
JButton cancelButton = new JButton("Cancel Ticket");
JButton clearButton = new JButton("Clear");
frame.setLayout(new GridLayout(11, 2, 10, 10));
frame.add(title);
frame.add(new JLabel());
frame.add(nameLabel);
frame.add(nameField);
frame.add(trainNoLabel);
frame.add(trainNoField);
frame.add(trainNameLabel);
frame.add(trainNameField);
frame.add(classLabel);
frame.add(classBox);
frame.add(dateLabel);
frame.add(dateField);
frame.add(sourceLabel);
frame.add(sourceField);
```

```
frame.add(destinationLabel);
frame.add(destinationField);

frame.add(bookButton);
frame.add(cancelButton);
frame.add(clearButton);
frame.add(new JLabel());
// BOOK TICKET
bookButton.addActionListener(e -> {
    String passengerName = nameField.getText().trim();
    String trainNumberText = trainNoField.getText().trim();
    String trainName = trainNameField.getText().trim();
    String journeyDate = dateField.getText().trim();
    String source = sourceField.getText().trim();
    String destination = destinationField.getText().trim();
    // Validation
    if (passengerName.isEmpty() ||
            trainNumberText.isEmpty() ||
            trainName.isEmpty() ||
            journeyDate.isEmpty() ||
            source.isEmpty() ||
            destination.isEmpty()) {
        JOptionPane.showMessageDialog(frame,
                "Please fill all fields!",
                "Validation Error",
                JOptionPane.ERROR_MESSAGE);
        return;
    }
```

```
int trainNumber;
try {
    trainNumber = Integer.parseInt(trainNumberText);
} catch (NumberFormatException ex) {
    JOptionPane.showMessageDialog(frame,
            "Train number must be numeric!",
            "Invalid Input",
            JOptionPane.ERROR_MESSAGE);
    return;
}
// Validate date
try {
    SimpleDateFormat format =
            new SimpleDateFormat("dd-MM-yyyy");
    format.setLenient(false);
    Date date = format.parse(journeyDate);
} catch (Exception ex) {
    JOptionPane.showMessageDialog(frame,
            "Enter date in DD-MM-YYYY format!",
            "Invalid Date",
            JOptionPane.ERROR_MESSAGE);
    return;
}
int pnr = generatePNR();
String sql = """
        INSERT INTO reservations
        (pnr, passenger_name, train_number, train_name,
         class_type, journey_date, source, destination)
        VALUES (?, ?, ?, ?, ?, ?, ?, ?)
```

```
        """;
    try (Connection con = getConnection();
         PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setInt(1, pnr);
        ps.setString(2, passengerName);
        ps.setInt(3, trainNumber);
        ps.setString(4, trainName);
        ps.setString(5, classBox.getSelectedItem().toString());
        ps.setString(6, journeyDate);
        ps.setString(7, source);
        ps.setString(8, destination);
        ps.executeUpdate();
        JOptionPane.showMessageDialog(frame,
                "BOOKING SUCCESSFUL!\n\n"
                        + "PNR Number: " + pnr
                        + "\nPassenger: " + passengerName
                        + "\nTrain: " + trainName
                        + "\nFrom: " + source
                        + "\nTo: " + destination
                        + "\nClass: "
                        + classBox.getSelectedItem(),
                "Booking Confirmation",
                JOptionPane.INFORMATION_MESSAGE);
    } catch (SQLException ex) {
        JOptionPane.showMessageDialog(frame,
                "Booking Error: " + ex.getMessage(),
                "Database Error",
                JOptionPane.ERROR_MESSAGE);
```

```
    }
});
// CANCEL TICKET
cancelButton.addActionListener(e -> {
    String input = JOptionPane.showInputDialog(
            frame,
            "Enter PNR Number:"
    );
    if (input == null || input.trim().isEmpty()) {
        return;
    }
    int pnr;
    try {
        pnr = Integer.parseInt(input);
    } catch (NumberFormatException ex) {

        JOptionPane.showMessageDialog(frame,
                "Invalid PNR number!");
        return;
    }
    String selectSQL =
            "SELECT * FROM reservations WHERE pnr = ?";
    try (Connection con = getConnection();
         PreparedStatement ps =
                 con.prepareStatement(selectSQL)) {
        ps.setInt(1, pnr);
        ResultSet rs = ps.executeQuery();
        if (rs.next()) {
            String details =
```

```
        "PNR: " + rs.getInt("pnr")
        + "\nPassenger: "
        + rs.getString("passenger_name")
        + "\nTrain: "
        + rs.getString("train_name")
        + "\nJourney Date: "
        + rs.getString("journey_date")
        + "\nFrom: "
        + rs.getString("source")
        + "\nTo: "
        + rs.getString("destination");
int choice = JOptionPane.showConfirmDialog(
        frame,
        details
                + "\n\nAre you sure you want to cancel?",
        "Booking Details",
        JOptionPane.YES_NO_OPTION
);
if (choice == JOptionPane.YES_OPTION) {
    String deleteSQL =
            "DELETE FROM reservations WHERE pnr = ?";
    try (PreparedStatement deletePS =
                 con.prepareStatement(deleteSQL)) {
        deletePS.setInt(1, pnr);
        deletePS.executeUpdate();
        JOptionPane.showMessageDialog(
                frame,
                "Booking Cancelled Successfully!"
        );
```

```
            }
        }
    } else {
        JOptionPane.showMessageDialog(
                frame,
                "No booking found for this PNR!",
                "Not Found",
                JOptionPane.WARNING_MESSAGE
        );
    }
} catch (SQLException ex) {
    JOptionPane.showMessageDialog(
            frame,
            "Database Error: " + ex.getMessage()
    );
}
});
// CLEAR BUTTON
clearButton.addActionListener(e -> {
    nameField.setText("");
    trainNoField.setText("");
    trainNameField.setText("");
    dateField.setText("");
    sourceField.setText("");
    destinationField.setText("");

    classBox.setSelectedIndex(0);
});
frame.setSize(650, 550);
```

```
        frame.setLocationRelativeTo(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
    // MAIN METHOD
    public static void main(String[] args) {
        createTable();

        SwingUtilities.invokeLater(() -> {
            loginWindow();
        });
    }
}
```

10. Sample Output

```
====================================
      ONLINE RESERVATION SYSTEM
====================================
Username: admin
Password: ****
Login Successful!

----- RESERVATION FORM -----
Passenger Name: Kowsalya
Train Number: 12601
Train Name: Chennai Express
Class: AC
Journey Date: 15-10-2026
Source: Coimbatore
```

Destination: Chennai

Booking Successful!

PNR Number: 458721

Passenger Name: Kowsalya

Train: Chennai Express

From: Coimbatore

To: Chennai

Class: AC

Ticket Confirmed!

11. Cancellation Output

Enter PNR Number: 458721

Booking Details Found!

Passenger Name: Kowsalya

Train Number: 12601

Train Name: Chennai Express

Are you sure you want to cancel? YES

Booking Cancelled Successfully!

12. Testing

Test

Input

Expected Result

Login

Correct username/password

Login successful

Login

Wrong password

Access denied

Passenger

Empty name

Validation message

Train number

Numeric value

Accepted

Booking

Valid details

Booking successful

PNR

Valid PNR

Booking details displayed

Cancellation

Confirm

Booking deleted

13. Demo Video

In your video first 2 seconds show:

Kowsalya K

Java Development

Online Reservation System

Then show the application working: Login → Reservation → Booking → PNR → Cancellation

LINK: https://drive.google.com/file/d/1CTrIpcfkKS9L1EW1UOOJ3o2Z6nPLNyQK/view?usp=drivesdk

15. Conclusion

The Online Reservation System provides a simple way to manage train reservations using Java. It demonstrates GUI development, database connectivity, user authentication, booking, PNR generation, cancellation and input validation.